# CSS LAYOUT ARCHITECTURE

## Grid vs. Flexbox Guide

## 1 OVERVIEW

Modern CSS provides two powerful layout systems. Choosing the right tool depends on whether your design problem is one-dimensional or two-dimensional.

## 2 HOW IT WORKS

### FLEXBOX (One-Dimensional)

- Best for a row or column.
- Content-driven (sizes based on content).
- Ideal for navbars and component alignment.

### CSS GRID (Two-Dimensional)

- Handles rows and columns.
- Layout-driven (defines grid tracks first).
- Ideal for page layouts, dashboards and complex structures.

## 3 CORE DIFFERENCE

| FLEXBOX | CSS GRID |
|---|---|
| One-dimensional | Two-dimensional |
| Row OR column | Rows AND columns |
| Content-driven | Layout-driven |
| Best for components | Best for full page layouts |

## 4 EXAMPLE: CHOOSING THE RIGHT LAYOUT

### FLEXBOX (1D)

Single row or single column



Row (horizontal)



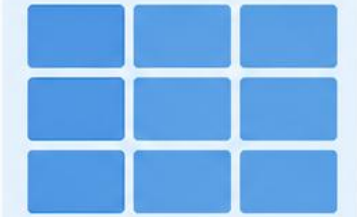
Column (vertical)

**FLEXBOX PROPERTIES**

```
display: flex;
flex-direction: row | column;
justify-content: space-between;
align-items: center;
flex-wrap: wrap;
```

### CSS GRID (2D)

Multiple rows and columns



**GRID PROPERTIES**

```
display: grid;
grid-template-columns: ...;
grid-template-rows: ...;
gap: 1rem;
grid-template-areas: ...;
```

### DECISION CHEAT SHEET

1. Are you aligning items along a single axis (row or column)?
   → USE FLEXBOX
2. Do you need to control row tracks and column tracks simultaneously?
   → USE CSS GRID
3. Can you use both together?
   **YES!** Grid handles the macro page layout, while Flexbox handles the micro component alignment inside individual grid cells.

## 5 LIMITATIONS

- ✓ Flexbox is not ideal for complex 2D layouts.
- ✓ Grid can be overkill for simple components.
- ✗ Both require modern browser support.
- ✗ Learning curve for advanced features.

## KEY TAKEAWAY

Use Flexbox for one-dimensional layouts, use Grid for two-dimensional layouts, and combine both when needed.

*"Master the axis, control the space."*